WORKED SOLUTION

STEP II 1991, Question 16

Probability and Statistics • Binomial Normal Approximation

STEP II 1991, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s2-q16

The key idea

Track the *deficit*: how far the water level is below the top of the dam. Outflow raises it steadily, and each shower lowers it by exactly V; overflow means the deficit goes negative. In the two-day problem only one pattern of rain can possibly cause overflow, and then just one uniform time matters. In the summer the outflow is switched off, so everything comes down to counting showers, and that count is binomial.

Solution

Part (i): the two-day holiday

Measure time in days from the midnight the holiday starts, and let $D(t)$ be the deficit at time t, so $D(0) = V$. Between showers D increases at rate u, and each shower decreases it by V. The dam overflows if D ever becomes negative. (We assume, as the question intends, that the reservoir holds enough water for the outflow to continue throughout the two days.)

Let R_1, R_2 be the events that it rains on day 1 and on day 2, and let the rain times (as fractions of a day) be T_1 and T_2, each uniform on $[0,1]$. The days are independent, as is implicit in "a probability p on any given day".

- **No rain:** the deficit only grows; no overflow.
- **Rain on one day only**, at time s: just after the shower $D = V + us - V = us \geq 0$, so no overflow (apart from the probability-zero event $s = 0$).
- **Rain on both days:** after the first shower $D = uT_1 \geq 0$, which is safe. Just after the second, at time $1 + T_2$,

$$D = V + u(1 + T_2) - 2V = u(1 + T_2) - V.$$

  Notice that T_1 has dropped out: the outflow runs at the same rate whenever the first shower falls.

So overflow happens exactly when both days are wet and $u(1+T_2) < V$, that is, $T_2 < \frac{V}{u} - 1$. If $\frac{1}{2}V \le u \le V$, the number $\frac{V}{u} - 1$ lies in $[0,1]$, and since T_2 is uniform,

$$\mathrm{P}(\text{overflow}) = p^2\left(\frac{V}{u} - 1\right).$$

Setting this equal to Q:

$$\frac{V}{u} = 1 + \frac{Q}{p^2} \quad \Longrightarrow \quad u = \frac{p^2 V}{p^2 + Q}.$$

The condition $Q < p^2$ is what makes this consistent: $0 < Q/p^2 < 1$ gives $\frac{1}{2}V < u < V$, as assumed. (Without it there's no solution. The overflow probability can never exceed p^2, because two wet days are needed.)

Part (i): the summer holiday

Now $u = 0$, so the deficit never increases. It starts at kV and drops by V with each shower, so if N is the number of wet days among the 18,

$$\text{overflow} \iff NV > kV \iff N > k.$$

The days are independent, each wet with probability $\frac{1}{3}$, so $N \sim \mathrm{B}(18, \frac{1}{3})$, with mean 6 and variance $18 \cdot \frac{1}{3} \cdot \frac{2}{3} = 4$.

> **Ask yourself** Which approximation is "suitable" here, and why do we need a continuity correction?

With $np = 6$ and $n(1-p) = 12$, the normal approximation $N \approx \mathrm{N}(6, 4)$ is reasonable. (A Poisson approximation isn't, because p isn't small.) N takes whole-number values, so for an integer k the event $N > k$ is the event $N \ge k+1$, which we approximate by $X > k + \frac{1}{2}$ with $X \sim \mathrm{N}(6,4)$. We need

$$\mathrm{P}\left(Z > \frac{k + \frac{1}{2} - 6}{2}\right) < \frac{1}{10} \iff \frac{k - 5.5}{2} > 1.2816 \iff k > 8.06.$$

Concretely, $k = 8$ gives $\mathrm{P}(Z > 1.25) \approx 0.106$, which is too big, and $k = 9$ gives $\mathrm{P}(Z > 1.75) \approx 0.040$, which is fine. Since overflow depends only on whether $N > k$, a non-integer k between 8 and 9 is no better than $k = 9$. So the engineer should drain the reservoir to

$$k = 9,$$

that is, $9V$ below the top.

Part (ii)

Now there are 36 half-days, independent, each wet with probability $\frac{1}{6}$. The outflow is still zero, so overflow again means $N > k$, where now $N \sim \mathrm{B}(36, \frac{1}{6})$. The mean is still 6, but the variance is $36 \cdot \frac{1}{6} \cdot \frac{5}{6} = 5$, which is larger than 4.

With the same mean, a larger variance means a heavier upper tail: for each k above the mean, $\mathrm{P}(N > k)$ is larger than before. So the required k is *higher*. Approximating by $\mathrm{N}(6, 5)$ with the continuity correction, we need

$$\frac{k - 5.5}{\sqrt{5}} > 1.2816 \iff k > 8.37,$$

compared with $k > 8.06$ before. (The normal approximation gives $\mathrm{P}(N \geq 9) \approx 0.132$ here, against 0.106 in part (i).) So the threshold rises. If k is rounded up to a whole number, both cases still give $k = 9$. But the tail at $k = 9$ is now about 0.06 rather than 0.04, so the margin of safety is thinner.

Checks

- A Monte Carlo simulation of the two-day holiday (400,000 runs) with $p = 0.5$, $Q = 0.1$ and $u = \frac{0.25}{0.35}V \approx 0.714V$ gave an overflow frequency of 0.0996. Other choices of p and Q agreed just as well.
- Limiting cases: as $Q \to p^2$ we get $u \to \frac{1}{2}V$, the slowest outflow at which two wet days can still cause overflow. As $Q \to 0$ we get $u \to V$, at which a second shower can never overflow.
- The exact binomial tails confirm the normal approximation. For $\mathrm{B}(18, \frac{1}{3})$, $\mathrm{P}(N \geq 9) = 0.1076$ and $\mathrm{P}(N \geq 10) = 0.0433$. For $\mathrm{B}(36, \frac{1}{6})$, $\mathrm{P}(N \geq 9) = 0.1332$ and $\mathrm{P}(N \geq 10) = 0.0652$.

Common slips

- Thinking the first rain time T_1 matters in the two-day problem. It cancels, because the outflow is constant.
- Forgetting the continuity correction. Without it the inequality becomes $k > 8.56$, which hides the fact that N is a whole number.
- Saying the answer to (ii) is "the same" because the means are equal. It's the spread that controls the tail.

Beyond STEP

Rain arriving at random instants, and water draining away at a steady rate, is a classic model in applied probability: storage theory, developed by Moran for dams. It is closely related to queues and insurance ruin. In part (ii), as the day is cut into more and smaller pieces with the mean held fixed, $\mathrm{B}(n, 6/n)$ tends to the Poisson distribution with mean 6 and variance 6, and the arrivals become a Poisson process. For more, see Grimmett and Welsh, *Probability: An Introduction*.

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